Contact Frictional Forces

- Friction: is an essential force for most types of motion. It is appear every day in our life. Without it we would not be able to walk, automobiles would not be able to move, and even machinery would not be able to function.
- It is the contact force acting parallel to the surface of contact (as contrasted with the normal force which is also a contact force but is directed perpendicular to the contact surface). It is produced by electromagnetic interactions between the molecules at the contacting surfaces of the two objects.
- The maximum force of friction F is:

$$F = \mu N$$

- N is a normal force, μ is the coefficient between the two surfaces.it depends upon the two materials in contact, and it is essentially independent of the surface area, as shown in Table

Example Values of Coefficient of Friction

Material'	μ (static **friction**)
Steel on steel	0.15
Rubber tire on dry concrete road	1.00
Rubber tire on wet concrete road	0.7
Steel on ice	0.03
Between tendon and sheath	0.013
Lubricated bone joint	0.03

- a frictional force between moving objects is always in a direction to slow the motion and is called sliding friction or kinetic friction. The frictional force on object is directed opposite to its velocity. It depends on the nature of the material surfaces, surprisingly, it does not depend on the contact area (to a good approximation).
- The kinetic friction is proportional to the normal force F_N, and can be written as:

$$F_{KFr} = \mu_k F_N$$

μ_k: is the coefficient of kinetic friction, which depends on the two material surfaces. This is clearly not a vector equation because F_{KFr}: is parallel and F_N is perpendicular to the surface.

This equation is an empirical approximate one and the coefficient of kinetic friction depends on the degree of smoothness of the surfaces, as well as on whether they are wet or lubricated.

- Always opposes the direction of motion (sliding).
- static friction: Two objects are in contact, but at rest with respect to each other, there are also, molecular bonds that form between contact points. Just sitting at rest does not result in any net force along a direction parallel to the contact surface; if there were, this force would spontaneously make the object accelerate. But if we try to push a object with a force directed along the table surface, the molecular bonds supply a frictional force in the opposite direction, opposing the impending motion.
- It arises in response to an applied force that would result in motion. As long as there is no motion, the static friction force is always as large as it has to be to cause a net balance of all forces on the object.
- Imagine applying a force to our block on the table, starting with a small force and increasing its strength gradually. Until the molecular bonds are ruptured to allow motion, the static friction is exactly equal and opposite to the applied force and there is no motion.

$$F_{SFr,max} = \mu_s F_N$$

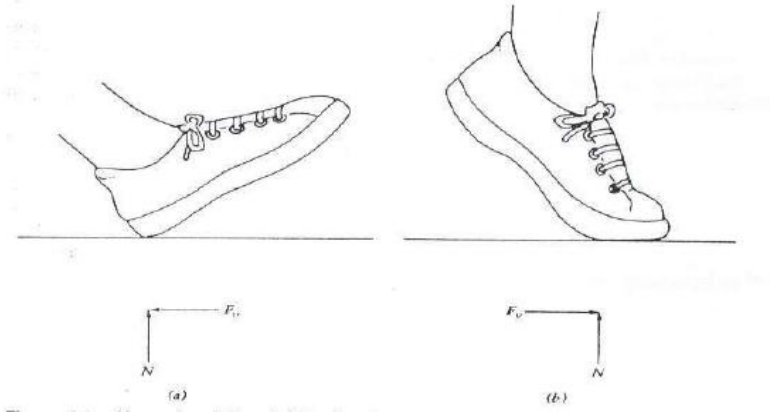
μ_s: is the coefficient of static friction,

- In general, the static friction force will be less than, or at most equal to $\mu_s F_N$

$$F_{SFr} \leq \mu_s F_N$$

- The normal force N is crucial and is determined by applying Newton's Second Law in the direction perpendicular to the surface.
- Example in biology:
  - Friction helps red blood cells grip capillary walls as they deform and squeeze through.
  - In joints, synovial fluid reduces friction, allowing smooth motion of bones.
- When a person is walking, as the heel of the foot touches the ground a force is transmitted from the foot to the ground.

- We can resolve this force into horizontal and vertical components. The vertical reaction force is applied by the surface and is labeled N (normal force).
- The horizontal reaction component must be applied by frictional forces, as shown in figure



- Normal walking. (a) Both a horizontal frictional componentof force F_N and a vertical (normal) component of force N exist on the heel as it strikes the ground. Friction between the heel and surface prevents the foot from slipping forward.

· (b) When the foot leaves the ground the frictional component of force F_N prevents the toe from slipping backward.

Measurements have been made of the horizontal force component of the heel as it strikes the ground when a person is walking, and it has been found to be = 0.15 W, where W is the person's weight.

The frictional force is large enough both when the heel touches down and when the toe leaves the surface to prevent a person from slipping.

This shows how large the frictional force must be in order to prevent the heel from slipping.

The coefficient of friction in bone joints is very small

If a disease of the joint exists, the friction may become large.

The synovial fluid in the joint is involved in the lubrication.

The saliva we add when we chew food acts as a lubricant (to reduce the friction force.(

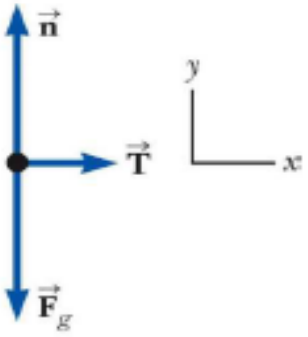
For example, if you swallow a piece of dry toast you become painfully aware of this lack of lubricant

The Object Under a Net Force, the dynamical force

According to second law of Newton, the force is equal:

F= ma = Δt

mv: momentum.



Example :Forces acting on the crate:

A tension, acting through the rope, is the magnitude of force$\vec{T}$.

The gravitational force $\overrightarrow{f_g}$

The normal force ,$\vec{n}$ exerted by the floor.

Apply Newton's Second Law in component form:

$$\sum f_x = T = ma_x$$

$$\sum f_y = n - F_g = 0 \rightarrow n = F_g$$

Solve for the unknown(s).

Circular Motion Dynamics

- An object moving in a circle at constant speed is undergoing an acceleration because its velocity vector is constantly changing direction
- A particle traveling in a circle at constant speed has an acceleration directed radially inward, toward the center of the circle, known as the centripetal ("center-seeking") acceleration.
- centripetal force: When a particle travel in uniform circular motion a net force must be applied to it in the direction of the centripetal acceleration.
- The net force that points toward the center of the circle. It must be provided by a real, physical force (or combination of forces) such as:

1: Tension (a ball on a string) 2: Friction (a car rounding a curve)

3: Gravity (a planet orbiting a star)

4: Normal Force (a roller coaster at the top of a loop)

1. Tension (a ball on a string): When you swing a ball on a string, the string pulls the ball inward. This tension force in the string is the centripetal force that constantly changes the ball's direction, keeping it in a circular path.
2. Gravity (a planet orbiting a star): The gravitational pull between a star and a planet is what keeps the planet in orbit. This force of gravity acts as the centripetal force, continuously pulling the planet toward the star and preventing it from flying off in a straight line.
3. Friction (a car rounding a curve): When a car goes around a curved road, friction between the tires and the road surface provides the necessary centripetal force. This frictional force pushes the car toward the center of the curve, allowing it to turn.
4. Normal Force (a roller coaster at the top of a loop): At the top of a roller coaster loop, the track pushes down on the car. This normal force (often combined with the car's weight) provides the centripetal force needed to keep the car moving in its circular path, even when it's upside down.

- When applied Newton's second law to an object traveling in uniform circular motion, acceleration is centripetal acceleration.

$$F_{net} = ma_{cent} = \frac{mv^2}{r}$$

m: mass, a_{cent} : Centripetal Acceleration$v$: constant speed r: Radius of the circle

- The net force must point toward the center of the circle.
- Example:
  - Electrons orbiting nuclei experience centripetal attraction.
  - In biology, similar principles apply in centrifugation and cell rotation.
- A biological mixture (such as a blood sample or a mixture of cells) is placed in special tubes inside the centrifuge machine. The machine spins these tubes at very high rotational speeds, which generates an immense centripetal force directed toward the center of the rotation. This strong centripetal force causes the denser components in the mixture (like red blood cells or cellular organelles) to sediment (move outward) faster and more forcefully than the less dense components (like the liquid plasma in blood). This process is called sedimentation.

- Importance and Use
  This process is vital for separating biological materials. Practical applications include:
  - Separating Blood Components: To isolate red blood cells, white blood cells, and platelets from blood plasma.
  - Isolating Cells or Organelles: To separate specific types of cells or internal cell components like mitochondria from a cell lysate.
  - Concentrating Samples: To pellet solid materials at the bottom of the tube after separating them from the liquid.

The Center of Mass (COM)

- The center of mass: is the average position of all the mass in a system.
- A system of (n) particles is defined to be the point whose coordinates are given by

$$\overrightarrow{r_{com}} = \frac{1}{M}\sum_{i=1}^{n} m_i \vec{r_i}$$

M: The total mass of the system. Which = $(\mathrm{m_1} + \mathrm{m_2} + \ldots\ldots\ldots\ldots\ldots\ldots\ \mathrm{m_n})$

$m_i =$ mass of (i) particle.

For a System of (n) Particles: The position of the center of mass in 1D

$$x_{com} = \frac{1}{M}\sum_{i=1}^{n} m_i x_i$$

$$= \frac{\mathrm{m_1x_1+m_2x_2+\ldots\ldots\ldots\ldots\ldots\ldots .m_nx_n}}{\mathrm{m_1+m_2+\ldots\ldots\ldots\ldots\ldots\ldots\ m_n}}$$

$$= \frac{\mathrm{m_1x_1+m_2x_2+\ldots\ldots\ldots\ldots\ldots\ldots\ m_nx_n}}{\mathrm{M}}$$

For 2D, and 3D is given by:

$$x_{com} + y_{com} = \frac{1}{M}\sum_{i=1}^{n} m_i x_i + \frac{1}{M}\sum_{i=1}^{n} m_i y_i = \frac{1}{M}\left(\sum_{i=1}^{n} m_i x_i \hat{\imath} + \sum_{i=1}^{n} m_i y_i \hat{\jmath}\right)$$

$$x_{com} + y_{com} + z_{com} = \frac{1}{M}\sum_{i=1}^{n} m_i x_i \hat{\imath} + \frac{1}{M}\sum_{i=1}^{n} m_i y_i \hat{\jmath} + \frac{1}{M}\sum_{i=1}^{n} m_i z_i \hat{k}$$

$$\frac{1}{M}\left(\sum_{i=1}^{n} m_i x_i \hat{\imath} + \sum_{i=1}^{n} m_i y_i \hat{\jmath} + \sum_{i=1}^{n} m_i z_i \hat{k}\right)$$

- For Solid Objects (Continuous Mass Distribution): We replace the sum with an integral:

$$\overrightarrow{\mathrm{r_{com}}} = \frac{1}{\mathrm{M}} \int \mathrm{r}\, \mathrm{d}m$$

- The center of mass moves as if all the mass of the system were concentrated at that point and all external forces were applied there.

- Motion of the Center of Mass:

  Velocity of COM: $v_{com} = \frac{m_1 v_1 + m_2 v_2 + \ldots\ldots\ldots\ldots\ldots\ldots m_n v_n}{M}$

  Acceleration of COM: $a_{com} = \frac{m_1 a_1 + m_2 a_2 + \ldots\ldots\ldots\ldots\ldots\ldots m_n a_n}{M}$

- The motion of the whole system can be analyzed as if all its mass were concentrated at the CM.
- Even in complex biological systems, like a running animal or a swimming fish — the center of mass follows the same physical laws of motion.
- Examples:
- A high jumper's body can clear the bar even if their CM passes under the bar due to body contortions.

The Physical Principle: A high jumper can successfully clear the bar even if their Center of Mass (CM) passes underneath the bar.
How is this possible? Through sophisticated body contortions. By arching their body, the jumper allows their limbs and head to clear the bar while their overall center of mass (the average position of all their body's mass) travels on a lower, more efficient path that can be below the bar itself. This is a clever application of mechanical principles.

- Maintaining balance involves keeping your center of mass over your base of support.

The Fundamental Principle of Balance: Maintaining balance involves keeping your center of mass directly over your base of support.
Base of Support: This is the area bounded by the points of contact with the ground (e.g., the soles of your feet and the area between them).
Application: If your center of mass moves outside this base, you become unstable and will likely fall. This principle is fundamental to all activities like walking, standing, and sports.

- Applications:
- The human body’s CM helps determine balance and posture.

The location of the CM is key to understanding balance and posture. Poor posture often means the CM is not optimally aligned, leading to strain on muscles and joints.

- Physical therapists and athletes study CM motion to improve movement efficiency.

Physical Therapists: Analyze CM motion to help patients regain proper balance and gait after injuries.

Athletes: Study CM motion to improve movement efficiency, enhance performance, and minimize wasted energy (e.g., perfecting running techniques, jumps, or weightlifting form).

- In microscopic systems, the CM concept helps describe cell motion and molecular assemblies.

The CM concept is also used to describe the motion of cells (how they move and migrate).
It helps in understanding how molecular assemblies form and behave.

- Newton's Second Law for a System:

$$\sum F_{ext} = Ma_{com}$$

- This is a profound result. It states that the net external force on a system determines the acceleration of its center of mass. Internal forces (forces between particles within the system) have no effect on the motion of the COM.

Momentum

- While energy describes how much work a system can do, momentum describes how motion is shared or transferred between interacting bodies.
- Linear Momentum (p):

Linear momentum is a vector quantity (direction is the same as the velocity vector). It has both magnitude and direction. It defined as the product of an object's mass and its velocity. And given by:

$\vec{P} = m\vec{v}$ SI Units: kg·m/s (which is equivalent to N·s).

- An object's momentum changes only when a net external force acts on it.
- Newton's Second Law in Terms of Momentum
- The original form of Newton's Second Law is actually, the net force on a particle is equal to the rate of change of its momentum).

$$\overrightarrow{F_{net}} = \frac{\mathrm{d}\vec{p}}{dt}$$

For a constant mass, this simplifies to the familiar:

$$\overrightarrow{F_{net}} = \frac{\mathrm{d}(m\overrightarrow{v})}{dt} = m\frac{\mathrm{d}\vec{v}}{dt} = m\,\vec{a}$$

- This form is more fundamental and is correct even in situations where mass changes (like a rocket propulsion (expelling fuel) or a biological system exchanging material.

- Rocket Propulsion: A rocket expels burnt fuel (mass) at high speeds backward, causing its own mass to continuously decrease. The thrust force on the rocket is calculated using this general form (F=$\frac{dp}{dt}$)
- Biological Systems: A living organism that gains or loses mass (like eating food or burning fat) while in motion.

- Example 1: A 60 kg person walking at 1 m/s bumps into a wall and stops in a distance of 2.5 cm in about 0.05 sec. What is the force developed on impact?

$$\Delta\,(mv) \;=\; (60\;\;kg)\;\;(1\,m/s\}) \;-\; (60\;\;kg)\,(0\;\;m/s) \;=\; 60\,kg\,m/s\}$$

The force developed on impact is:

$$F \;=\frac{\Delta(mv)}{\Delta\,t} = \;\frac{60}{0.05}\;\;= 1200\,kg\,m/s^2$$

F = 1200 N

Example 2: A 50 kg person jumping from a height of 1 m is traveling at 4.5 m/s just prior to landing. Suppose she lands on a pad and stops in 0.2 sec. What maximum force will she experience?

$$F \;=\frac{\Delta(mv)}{\Delta\,t} = \frac{50 * 4.5}{0.2} = 1125\,N$$

Example 3: Estimate the force on the forehead if the mass of the head is 4 kg, its velocity is 15 m/s, and the padded dash stops it in 0.002 sec.

$$F \;=\frac{\Delta(mv)}{\Delta\,t} = \frac{4 * 15}{0.002} = 3 * 10^4 10\,N$$

Impulse

- When a force acts on an object for a period of time, it changes the object's momentum. This change is called impulse. It is a measure of the change in momentum. It is a vector quantity.
- For a Constant Force:

$$J = F_{avg}\Delta t$$

- F_{avg} : Is the average net force and, Δt: Is the time interval over which it acts.
- · For a Variable Force:

$$j = \int F\,(\mathrm{t})dt$$

- The impulse is the integral of the force over the time interval.
- Example in biology:
  - When you catch a ball, your hands apply a force over time to stop it.

- Extending the time (by moving your hands backward) reduces the peak force — this is why athletes “give” with the ball.
- The same principle protects organs and joints from shock.

The Impulse-Momentum Theorem

$$J = \Delta p$$

- This theorem is always true and is derived from $\sum F = \frac{dp}{dt}$. It provides a direct connection between the force applied over a time interval and the resulting change in velocity. It's especially useful for analyzing collisions where forces are large and time intervals are short,
- Cushioning a Blow: To reduce the average force (F_{avg}) experienced during a change in momentum (Δp), you must increase the time (Δt) over which the change occurs.

Examples:

- Airbags in cars: They increase the time it takes for a passenger to stop during a crash, thereby significantly reducing the force exerted on their body.
- Padded floors in gyms: They increase the duration of the impact when feet hit the ground, reducing the force on joints.
- Bending your knees when landing from a jump: This increases the time over which your body comes to a stop, lessening the force on your legs and spine.
- A baseball player "giving" with the catch: By moving the glove backward upon catching the ball, the player increases the time to stop the ball, which reduces the force felt on their hand.
- All these actions increase \Delta t, thereby decreasing the damaging force (F_{avg}).

- Biological Context:
  The theorem explains the biomechanics of walking, running, and jumping.
  The Principle: The force your legs exert over the time of contact with the ground determines the change in your momentum.
- Practical Implications:
  - Walking and Running: Your muscles control the duration and magnitude of the push against the ground to change your body's momentum.

- Jumping: You extend the push-off time against the ground to achieve a higher jump.
- Changing Direction: You control the force and its duration to change your direction of motion effectively

Conservation of Linear Momentum

- If the net external force on a system is zero, then the total linear momentum of the system is constant (conserved).

$$\sum F_{ext} = 0,\ then\ P_{total,i} = P_{total,f}$$

- Why it Works: Internal forces always occur in action-reaction pairs (Newton's 3rd Law). These pairs are equal in magnitude and opposite in direction, and they act for the same time. Therefore, their impulses cancel out, causing no net change in the total momentum of the system.
- Biological examples:
  - When a fish expels water backward, it moves forward, total momentum is conserved between fish and water.
  - In locomotion, animals push backward on the ground, water, or air to propel themselves forward.
  - Inside cells, molecular motors like kinesin “walk” along filaments by transferring momentum to their environment.

Applications of Momentum Conservation

1. Collisions:

- A collision is an interaction where particles exert strong forces on each other for a relatively short time. Total linear momentum is always conserved in any collision (since internal forces cancel out
- Types of Collisions
- Elastic Collision: Both momentum and kinetic energy are conserved. (e.g., two billiard balls colliding, atoms colliding, gas molecule collisions).

$$\sum \vec{p}_{initial} = \sum \vec{p}_{final}$$

$$\sum K.E_{initial} = \sum K.E_{final}$$

- Inelastic Collision: Momentum is conserved, but kinetic energy is not. Some energy is converted to heat, sound, or deformation. Example: a lump of clay sticking to a wall.

$$\sum \vec{p}_{initial} = \sum \vec{p}_{final}$$

$$\sum K.E_{initial} > \sum K.E_{final}$$

- Perfectly Inelastic Collision: · A special case of inelastic collision, the objects stick together after the collision. after the collision and move with a common velocity. This results in the maximum possible loss of kinetic energy.

In biology, collisions are often inelastic — energy transforms into heat or structural changes.

Example:

- Red blood cells colliding and deforming in capillaries.
- Molecular collisions leading to chemical reactions.

- 2. Explosions: An object breaks apart into multiple fragments due to internal forces. The initial momentum is zero (if the object was at rest). After the explosion, the vector sum of the momenta of all fragments must still be zero. Fragments fly apart in such a way that their center of mass remains stationary.
- 3. Recoil: When a system expels mass in one direction, it moves in the opposite direction, a direct result of momentum conservation.
  - When a person fires a gun, the forward momentum of the bullet is equal and opposite to the backward momentum of the gun (recoil). The system (gun + bullet) has no net external force horizontally, so its total horizontal momentum remains zero.
  - Examples:
    - Squids and jellyfish move by jet propulsion — pushing water backward.
    - Human breathing: when air is exhaled, the body experiences a tiny backward recoil.
    - DNA ejection from viruses involves recoil — as DNA leaves the capsid, the remaining structure recoils slightly.
  - Biological Context:
    - Swimming and Flying: Animals propel themselves by pushing water or air backward (applying a force over time, imparting momentum to the fluid). By conservation of momentum, the animal moves forward.
    - Molecular Collisions: The motion and collisions of molecules within a cell are governed by momentum conservation.

- Biomechanics of a Jump: When you jump, you push down on the Earth. The Earth acquires a tiny, imperceptible amount of downward momentum, while you acquire an equal amount of upward momentum.

Momentum and Energy Compared

Concept Equation	**Conserved?**	**Type**	**Example**
Momentum	Always (if no external force)	Vector	Collisions, propulsion
Energy	Only if forces are conservative	Scalar	Work, heat, reactions

- Momentum depends on direction, while energy depends on speed and configuration.
- Both are essential for describing motion, but momentum conservation holds even when energy converts into other forms.